



Engenharia Informática
Introduction to Machine Learning

Deliverable 1

Projeto IAA

Automated Resource Bottleneck Identification in Cloud
Infrastructure via Multi-Class Classification

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*This project is conducted by three students enrolled in the same practical class,
adhering to the required group composition.*

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Context and Motivation

Modern cloud environments host heterogeneous workloads that demand high availability and strict Service Level Agreements (SLAs). When performance degrades, identifying whether the root cause is CPU saturation, memory exhaustion, disk I/O connection, or network congestion – the “binding constraint” – is critical for rapid remediation.

Currently, this diagnosis relies on manual dashboard monitoring by human operators, a method that is unscalable, prone to error, and slow during high-stress events. By leveraging historical telemetry (CPU, memory, disk, and network metrics) from controlled stress tests where ground truth is known, we can train a machine learning model to automate this diagnostic process. This shifts the operational paradigm from reactive manual inspection to proactive, data-driven identification.

Dataset Description

Dataset Source and Features

- **Primary Source:** The Numenta Anomaly Benchmark (NAB), specifically the server metrics subset containing real-world AWS EC2 instance telemetry.
<https://github.com/numenta/NAB>
- **Alternative Source:** The Alibaba Cluster Trace is being evaluated for large-scale production job telemetry.
<https://github.com/alibaba/clusterdata>
- **Raw Features:** The model utilizes four primary metric streams sampled at regular intervals: CPU utilization (%), Memory utilization (%), Disk throughput (MB/s), and Network I/O (MB/s).

Label Engineering and Origin

As explicit bottleneck labels are often missing in raw telemetry, we will implement a threshold-based labeling strategy. For instance, sustained CPU usage above 85% for N consecutive samples will be labeled “CPU-bound”. While this introduces subjectivity, it is a documented approach in infrastructure workload characterization.

Limitations and Potential Bias

- **Environment Specificity:** The NAB dataset originates from AWS; usage patterns may encode infrastructure-specific configurations that do not generalize to on-premises environments.
- **Temporal Autocorrelation:** The time-series nature of the data violates the independent and identically distributed (i.i.d.) assumption, requiring non-random splitting to prevent data leakage.
- **Subjectivity of Ground Truth:** Human-defined thresholds for labels may misrepresent complex, multi-resource bottlenecks.

Type of Machine Learning Problem

We frame this as a **supervised multi-class classification problem**.

- **Input (X):** A multidimensional feature vector derived from sliding windows of raw metrics. To capture temporal dynamics, we calculate the mean, standard deviation, and rate of change for each metric within the window.
- **Output (y):** A categorical label from five mutually exclusive classes: Normal, CPU-bound, Memory-bound, I/O-bound, and Network-bound.

Proposed Models: We will implement and compare **Random Forest** and **XGBoost**. These models are selected for their high performance on tabular data and their ability to provide feature importance scores, which aids in the “critical analysis” of which metrics most strongly signal a bottleneck.

Initial Risks and Assumptions

- **Assumption of Proxy Labels:** We assume that threshold-derived labels are a sufficiently accurate proxy for true operational bottlenecks.
- **Assumption of Feature Sufficiency:** We assume that statistical aggregates (mean/std) over fixed windows capture enough “state” to distinguish between different resource stresses.
- **Risk of Class Imbalance:** In real-world data, “Normal” states significantly outnumber bottleneck states. We must use F1-score and Macro-averaged Precision/Recall to ensure the model actually identifies rare bottleneck events.

- **Risk of Temporal Leakage:** Standard random shuffling may allow the model to “peek” into the future, leading to over-optimistic performance metrics that fail in real-time deployment.

Acknowledgment of AI Tools

In accordance with the project guidelines, we acknowledge the use of Claude Code to facilitate conceptual synthesis and provide an architectural overview of the problem domain, and Gemini to assist in the structural organization and formatting of this document.